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Quantifying methane emissions under field conditions under 2 different dairy production scenarios: Low-input versus high-input milk production

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ABSTRACT

Livestock production systems with ruminants play a relevant role in the emission of the greenhouse gas CH_4, which is known to significantly contribute to global warming. Consequently, it is a major societal concern to develop strategies in mitigating such emissions. In addition to breeding toward low-emitting cows, management strategies could also help in reducing greenhouse gas emissions from dairy farms. However, information is required for appropriate decision making. To the best of our knowledge, this is the first study that considers different, already available equations to estimate CH_4 emissions of small-scale dairy farms in the mountain region, which largely differ from large dairy farms in the lowlands concerning management and production. For this study, 2 different production systems, both typical for small-scale dairy farming in mountain regions, were simultaneously run over 3 yr at an experimental farm as follows: (1) a high-input production system, characterized by intensive feeding with high amounts of external concentrates and maize silage, year-round housing, and high yielding Simmental cattle breed, and (2) a low-input production system, characterized by prevailing hay and pasture feeding and silage ban, thus covering most of the energy requirements by forage harvested on-farm and the use of the local Tyrolean Grey cattle breed. Results reveal that feeding management has a significant effect on the amount of CH_4 emissions. The low-input production system produced less CH_4 per cow and per day compared with the high-input production system. However, if calculated per kilogram of milk, the high-input scenario produced proportionally less CH_4 than the low-input one. Findings of this study highlight the potential to assess in a fast and cost-effective way the CH_4 emission in different dairy production systems. This information contributes to the debate about the future of sustainable milk production in mountain regions, where the production of feed resources is climatically constrained, and could be useful for breeding purposes toward lower CH_4-emissions.
Key words: methane emission, dairy production, sustainability, Simmental, breeding

INTRODUCTION

Methane (CH_4), carbon dioxide (CO_2), and nitrous oxide (N_2O) are known to be some of the most relevant greenhouse gases and, thus, important drivers for global climate change (Schmithausen et al., 2018). Although CO_2 has a much longer lifetime in the atmosphere than CH_4, the latter has a many-fold higher global warming effect than that of CO_2 when present in the atmosphere (Boadi et al., 2004). Approximately 40% of the anthropogenic CH_4 production can be attributed to livestock production (Broucek, 2014; Aguirre-Villegas et al., 2015; Hou et al., 2015). The largest fraction of CH_4 produced by livestock farming originates from microbial fermentation of cellulosic feed material inside the rumen or, to a smaller extent, in the intestine, whereas a minor faction is formed during the decomposition of manure (Beauchemin et al., 2009; Garnsworthy et al., 2012; Broucek, 2014; Hou et al., 2015). The latter depends significantly on the animal housing system, as well as on manure storage and application systems, and can be reduced to a negligible amount when meeting optimal manure management strategies (Garnsworthy et al., 2012; Hristov et al., 2013). This could be reached by reducing, in general, the amount of liquid and stored manure (VanderZaag et al., 2014), which could be achieved, for instance, by increasing pasture access (Groenestein et al., 2012). Enteric CH_4 emissions, however, are more restricted in their management options as they primarily depend on the DMI, as well as on the feeding ration composition and on microbial fauna inside the digestive tract of ruminants (Ibidhi and Calsamiglia, 2020). Therefore, it is of great interest to assess enteric CH_4 emissions and, consequently,

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to develop strategies such as genetic selection for the permanent reduction of CH_4 production by ruminants (e.g., de Haas et al., 2021). Yet, direct measurement methods using, for example, respiration chambers or in vitro gas production techniques are highly cost-intensive and can only be applied with a limited number of animals, making it difficult to obtain relevant data on the whole breeding population (Engelke et al., 2018; Zhao et al., 2020). Thus, due to these restrictions, developing estimation formulas that can be applied for quantifying CH_4 production with high accuracy on a larger scale has been regarded as a major aim. Indeed, many authors, such as Kiggundu et al. (2020) and Eugène et al. (2019), performed enteric CH_4 emission estimations following the Tier 2 and Tier 3 equations, issued by the Intergovernmental Panel on Climate Change in 2006. These widely used guidelines are based on a CH_4 conversion factor (Y_m), that describes the percentage of gross energy in feed converted to CH_4 and relies on previous measurements made in respiration chambers (Storm et al., 2012). Nevertheless, these numbers are based on relatively generalized estimations and, therefore, might possibly affect the reliability of the results. With the objective to obtain more accurate values, Ramin and Huhtanen (2012) as well as Mills et al. (2003) have developed formulas based on parameters that can be measured or calculated directly, such as the DMI of cattle. Such variables, however, are not continuously and conveniently assessed in practice on dairy farms. Considering these restrictions, the necessity to provide a formula that takes into consideration variables that are routinely measured in practice (for instance, via milk recording) becomes evident. Therefore, Engelke et al. (2018) developed a formula with the aim to predict CH_4 emission based on ECM yield and milk mid-infrared spectra providing an estimate of milk fatty acid (**FA**) concentrations.

The aim of our study was to estimate/characterize for the very first time the CH_4 emissions of small-scale dairy farms in mountain region with contrasting production systems. For this purpose, we considered a low-input production system (roughage feed and pasture-based) with the autochthonous cattle breed Tyrolean Grey and a high-input production system (intensive use of external concentrates and silage, and year-round housing) with the high yielding Simmental cattle breed, simultaneously run at an experimental farm (Mair am Hof, Dietenheim, South Tyrol, Italy) by using already established equations considering, depending on the equation, routinely and experimentally collected production parameters. Both systems are practiced in mountain regions and have various effects on environmental and production traits (Katzenberger et al., 2020; Sabia et al., 2020; Zanon et al., 2021). Furthermore, the effect of stage of lactation and parity on the CH_4 emission was investigated. Differences between the 2 investigated systems were also expected because of the different cattle breeds adopted, as well as between the first and following lactations within the single breeds, as Islam et al. (2021) and Silvestre et al. (2021) have revealed that there are significant differences in CH_4 production strongly depending on cattle breed, mainly due to their microbial composition inside the rumen.

MATERIALS AND METHODS

The experimental and notification procedures were carried out in compliance with the European Union Directive 2010/63/EU.

Farm and Study Design

This study is part of the Action Plan 2016 to 2022 for Research and Training in the Fields of Mountain Agriculture and Food Science of the Autonomous Province of Bolzano/Bozen (Italy) started in 2019. Data from February 2019 to January 2022 were used for this study. Within this project, high- and low-input dairy cattle farming systems, both commonly found in the Alpine province of South Tyrol (Northeast Italy), are compared, focusing on various parameters such as animal health and welfare, economic rentability, and ecological footprint.

The experiment took place at the experimental farm Mair am Hof (46° 48′06.9″ N, 11°57′30.6″ E, 909 m above sea level; mean annual temperature 8.3°C and mean precipitation sum 977 mm/yr for 2019 to 2021; Teodone/Dietenheim, Val Pusteria/Puster Valley, South Tyrol, Italy). The low-input strategy is characterized by an extensive management system, following the hay-milk production scheme (EU, 2016), aiming to cover the majority of energy requirements by forage (hay and pasture feed), complying with a silage ban, and using the autochthonous Tyrolean Grey cattle breed (n = 15). The stocking method was a compartmented short sward grazing (German: *Kurzrasenweide*; Höllrigl et al., 2019) with 4 adjoining paddocks of 1.4 ha each (i.e., a continuously stocked pasture with stocking rate adjusted by means of restriction or enlargement of the grazed area, which is the number of paddocks used weekly), to achieve a target compressed sward height of 6 to 7 cm. The sward height was measured weekly by rising plate meter (Grasshopper G2 Sensor, App version 4.02, TrueNorth Technologies). During the grazing season (March–November), the animals had ad libitum pasture access and a maximum indoor feed integration

of about 37% of DMI of the total ration amount offered indoor during wintertime, with the pasture representing a large amount of the diet within this period (approximately 63% of DMI). The high-input system, on the other hand, is characterized by year-round housing of the animals and a feed ration mainly composed of maize silage, grass silage, and concentrates with the objective to obtain high milk yield using the high yielding Simmental cattle breed (n = 15). For both systems, a dual-purpose cattle breed was chosen, as Zanon et al. (2020) revealed the future economic potential of such breeds for Alpine dairy production systems.

Individual milk yield as well as energy uptake inside the stable have been routinely collected for both herds. The individual indoor DMI was continuously recorded by means of roughage intake control (RIC) feed-weigh troughs (Hokofarm Group). Forage analyses of all ration components were routinely performed at each variation of the feed ration, allowing computation of DM content and energy content of the ration in terms of NE_L according to RAP (2015). The daily milk production was measured by means of a milking parlor equipped with an electronic milk-recording device (Westfalia Dairy Plan, Westfalia-Surge). The herbage intake on the pasture was estimated by dividing the difference between the energy requirement on pasture and the energy intake in the barn by the energy content of the herbage on the pasture. The energy requirements on the pasture were estimated according to Macoon et al. (2003) with the following adjustments and assumptions: (1) the energy requirements for maintenance were assessed according to Kirchgeßner et al. (2014); (2) the energy requirements for BW changes were obtained by linear interpolation between 2 successive BW measurements. The latter were obtained by individual measurements of the lactating animals using a field scale (EziWeigh6i, Datamars Livestock), synchronized with the routine milk performance tests and carried out on average every 40 d. For the days preceding the first measurement after calving and those following the last measurement before calving, the slope of the following measurement interval or the previous measurement interval were respectively used; (3) the walked distance, according to an educated guess of the farm personnel, was set equal to 8 times the distance of the stall from the centroid of the paddocks (143.4 × 8 = 1,147.2 m), and the resulting energy expenditures were doubled according to Di Marco and Aello (2001), based on the mean slope of the paddocks (15.3% on average); and (4) the grazing time was set to 7 h/d.

The energy content of the herbage from the pasture was surveyed in each paddock according to a simplified Corrall-Fenlon method (Corrall and Fenlon, 1978), as modified by Mosimann (2001), and linearly interpolated between sampling dates. Milk quality was characterized during the routine milk performance tests carried out on average every 40 d, including the FA profile, by means of mid-infrared spectroscopy (Milko-Scan FT7, Foss Electric). All other parameters were related to these measurement dates as monthly mean values for each animal.

In addition to that, dairy cows were assigned to 2 groups based on parity (i.e., primiparous and multiparous cows). The lactation stage was expressed as lactation day at the time of the milk performance tests, whereas seasonality was accounted as week of the year for.

For better visualization of the differences between the investigated equations, calculations have been made, making use of 2 different quantification units of CH_4 production as follows: liter or megajoule of CH_4 produced per day and CH_4 emissions (liter or MJ) produced per kilogram of milk.

Feed Ration Composition. In Table 1, the 2 different feeding rations are summarized. The Simmental cattle was fed with a pre-defined feeding ration, slightly adapted over time, consisting on average of 12.9% hay (from different cuts), 25.3% maize silage, 25.8% grass silage, 34.9% concentrates, and 0.9% mineral feed. The Tyrolean Grey group, on the contrary, was fed with a ration containing 76.5% of hay (from different cuts), 21.3% concentrates, and 2.1% mineral feed (Table 1). During the vegetation period, the indoor DMI of the low-input group decreased (from 16.4 kg of average actual DMI inside the stable during the winter season to 6.6 kg during the grazing season), whereas DMI from pasture increased and accounted for 63.1% (11.3 kg) of total DMI (Table 1). Data for pasture intake could not be measured directly and was thus quantified as described above. On a yearly basis, pasture accounts for approximately one third of total DMI intake of the low-input group. In addition to the individually recorded DMI, the cows received a little amount of concentrates (0.5–1 kg/d) as a pet bait during the milking process, which the total DMI does not account for (Table 1).

Estimation of CH_4 Emissions. For quantifying CH_4 emission several previous published equations were considered, which use routinely as well as not routinely collected parameters. The equations were selected according to the availability of parameters recorded within our study as well as by the production environment under which they were developed, to generate a high accuracy of estimate (Table 2). The equations were as follows.

Ramin and Huhtanen (2012) predicted the CH_4 yield in L/d with the daily DMI as the variable in the following equation (**RH**):

Table 1. Mean daily DMI (average of monthly means of all available measurements) of the different components of the ration for Simmental and Tyrolean Grey dairy cattle

	Farming system					
	High-input		Low-input			
	All year		Winter season[1]		Grazing season[2]	
Ration component	Daily ration (kg of DM/d)	Proportion in the ration (%)	Daily ration (kg of DM/d)	Proportion in the ration (%)	Daily ration (kg of DM/d)	Proportion in the ration (%)
Hay	2.8	12.9	12.5	76.5	4.9	74.7
Grass silage	5.4	25.3	0.0	0.0	0.0	0.0
Maize silage	5.6	25.8	0.0	0.0	0.0	0.0
Concentrates	7.5	34.9	3.5	21.3	1.6	23.9
Mineral feed	0.2	0.9	0.3	2.1	0.1	1.4
Estimated herbage intake on pasture	0.0	0.0	0.0	0.0	11.3	63.1
Feed intake barn[3]	21.5	100.0	16.4	100.0	6.6	36.9
Total intake	21.5	100.0	16.4	100.0	17.9	100.0

[1]November–March (depending on the effective start date and end date of grazing).
[2]March–November (depending on the effective start date and end date of grazing).
[3]Average of the effective daily DMI recorded in the stable.

$$\text{RH: } CH_4 \text{ (L/d)} = 51.5 \times \text{DMI}^{0.792}.$$

Similarly, the equation of Engelke et al. (2018; **ENG1**) considers DMI as well as CH_4 yield in L/d as calculation units in the following equation:

$$\text{ENG1: } CH_4 \text{ (L/d)} = 361.4 + 18.9 \times \text{DMI} + 28.5 \times \text{C18:0} - 23.6 \times \text{C18:1 } \textit{cis},$$

where C18:0 and C18:1 *cis* are the stearic and vaccenic acid, respectively, expressed in % of the sum of all milk FA.

Mills et al. (2003) as well as Niu et al. (2021) consider the DMI for calculation, where CH_4 yield is indicated as megajoules per day in the following equations (**MILLS** and **NIU**, respectively):

$$\text{MILLS: } CH_4 \text{ (MJ/d)} = 56.27 - 56.27 \times e\ (-0.028 \times \text{DMI}),$$

$$\text{NIU: } CH_4 \text{ (MJ/d)} = 4.92 + 1.13 \times \text{DMI} - 0.118 \times \text{FA}.$$

The second equation published by Engelke et al. (2018; **ENG2**), on the contrary, is based on the ECM (calculated as indicated above) as well as on the SFA content of the milk, expressed in % of the sum of FA as follows:

$$\text{ENG2: } CH_4 \text{ (L/d)} = -1{,}364 + 9.58 \times \text{ECM} + 18.5 \times \text{SFA} + 32.4 \times \text{C18:0}.$$

Table 2. Investigation fields (cattle breed, housing system, ration) within publications

	Reference				
Item	Ramin and Huhtanen (2012)	Niu et al. (2021)	Engelke et al. (2018)	Mills et al. (2003)	Yan et al. (2006)
Cattle breed	—	Various breeds (mainly Holstein Friesian and Brown Swiss)	Holstein-Friesian	—	Holstein-Friesian
Housing system	58 freestall systems; 7 pasture systems; 18 tiestall barns	Various housing systems depending on the feeding experiment	Tiestall barns	Various housing systems	—
Ration	—	Diverse feeding systems, depending on publication: grass-, maize- and legume-based silage; concentrate mixture	Grass- and corn silage-based diet; Concentrate mixture	Maize-and grass silage, dried alfalfa; concentrate mixture	Grass silage-based diet; concentrate mixture

Yan et al. (2006) proposed the following 2 formulas, both accounting for DMI (kg/d; **YAN1**), whereas just one of them also considers the BW in kilograms (**YAN2**):

$$\text{YAN1: } CH_4 \text{ (L/d)} = 47.8 \times \text{DMI} - 0.76 \times \text{DMI}^2 - 4,$$

$$\text{YAN2: } CH_4 \text{ (L/d)} = 0.34 \times \text{BW} + 19.7 \times \text{DMI} + 12.$$

Taking into consideration the variables used to develop the equations, most DMI observations of the present study were found to be quite well covered by the data range of the other studies (Table 3). One exception was given by the high-input system in ENG1, with about 40% of the values beyond the upper limit and DMI values exceeding it by up to 7.5 kg/d (Supplemental Table SM1, https://data.mendeley.com/datasets/5wpdvx2vcm/1; Peratoner et al., 2023). In the same equation, the low-input system had about half of the observations of C18:0 lying below the lower limit, whereas this happened for the large majority of those of the high-input system (83%). On the contrary, the C18:1 *cis* values were fairly well covered by the range used to develop ENG1. Concerning ENG2, the ECM observations of the high-input system showed the same pattern observed for DMI in ENG1, whereas SFA exhibited a good agreement for both farming systems. Finally, concerning BW (accounted for in YAN2), a large proportion (57%) of BW values exceeding the upper limit of those used to develop the equation were found. All in all, all equations making use of DMI alone provides a good matching concerning the data range, whereas partial mismatching occurs for part of the variables (ECM, C18:0, BW) combined with at least one being well matched, with a higher matching deficit for the high-input system than for the low-input one but no clear suitability or unsuitability for just 1 of the 2 systems.

Statistical Analysis

The analysis of the estimated CH_4 emissions was performed by means of stepwise forward developed linear mixed models, starting from a baseline model accounting for the farming system (low-input/high-input) as a fixed factor and for the year and its interaction with the system as random terms. Values related to the same animal over time were treated as repeated measurements over the sequence of the measurement events (ordinally scaled) with the animal as a subject. The covariance structure providing the best fit was chosen using the Akaike's information criterion as an indicator. The usefulness of including further

Table 3. Percent of the observations of the present study for the high-input (n = 346) and the low-input group (n = 332) being lower than the minimum value (<min) or higher than the maximum value (>max) observed in the respective study to develop the equations[1]

			DMI (kg/d)		C18:0 (% of total fat)		C18:1 *cis* (% of total fat)		ECM (kg/d)		SFA (% of total fat)		BW (kg)	
Reference	Equation[2]	Farming system	<min	>max	<min	>max	<min	>max	<min	>max	<min	>max	<min	>max
Engelke et al., 2018	ENG1	High-input	0	37	83	0	3	1						
		Low-input	0	5	47	0	1	2						
Engelke et al., 2018	ENG2	High-input							0	37	1	0		
		Low-input							4	0	4	0		
Ramin and Huhtanen 2012	RH	High-input	0	4										
		Low-input	0	0										
Mills et al., 2003	MILLS	High-input	0	1										
		Low-input	4	0										
Niu et al., 2018	NIU	High-input	0	5										
		Low-input	0	0										
Yan et al., 2006	YAN1	High-input	0	11										
		Low-input	0	0										
Yan et al., 2006	YAN2	High-input	0	11									0	57
		Low-input	0	0									0	0

[1]See SM1 (Supplemental Table SM1, https://data.mendeley.com/datasets/5wpdvx2vcm/1; Peratoner et al., 2023) for details about the absolute values of the ranges.
[2]ENG1 = first equation by Engelke et al. (2018); RH = equation by Ramin and Huhtanen (2012); ENG2 = second equation by Engelke et al. (2018); MILLS = equation by Mills et al. (2003); NIU = equation by Niu et al. (2021); YAN1 = first equation by Yan et al. (2006); YAN2 = second equation by Yan et al. (2006).

Table 4. Mean overall measurements obtained at the milk performance tests or referred to the same dates ± SD of the independent variables used in the equations to estimate CH_4 emission depending on the farming system[1]

	Farming system		
		Low-input	
Parameter	High-input	Rest of the year	Grazing season
DMI (kg of DM/cow per d)	21.3 ± 3.03	16.4 ± 2.16	17.9 ± 2.80
ECM (kg/cow per d)	34.0 ± 8.37	22.2 ± 5.89	20.0 ± 5.10
SFA (g/100 g of FA)	65.1 ± 2.84	64.0 ± 2.64	59.9 ± 3.32
C18:0 (g/100 g of FA)	7.8 ± 1.27	8.2 ± 1.34	9.4 ± 1.49
C18:1 *cis* (g/100 g of FA)	18.7 ± 2.89	18.9 ± 3.30	22.6 ± 4.18
BW (kg)	761.2 ± 80.07	602.8 ± 58.18	576.9 ± 59.19

[1]For the low-input system, values according to the occurrence or absence of grazing are shown.

explanatory variables as well as their interaction with the system was stepwise tested using maximum likelihood as the estimation method in combination with the Satterthwaite approximation of the degrees of freedom, Akaike's information criterion as indicator to identify the variable to be added next, and the polynomial degree of the metric independent variables. The following variables were considered for inclusion into the statistical predictive model: parity (primiparous/multiparous) and the 2 metric variables, treated as covariates, seasonality (week of the year), and lactation stage (lactation day). The final model was computed using REML as the estimation method. Normality of residuals and homoscedasticity were checked by means of diagnostic plots, and data transformation was performed if necessary to meet these requirements. In these cases, back-transformed values are shown.

The correlation between CH_4 emissions estimated with different equations was explored by a Pearson test.

RESULTS AND DISCUSSION

Overview of the Independent Variables Used in the Equations

Most of the variables used to estimate the CH_4 emissions according to the different equations showed a clear differentiation depending on the farming system (Table 4). The high-input system resulted in higher DMI and ECM (+27% and 65%, respectively; Table 1). The different values in DMI between winter and grazing season for the low-input system are due to the estimated herbage intake on pasture. Moreover, the Simmental cattle group exhibited a higher BW (+30%) in comparison to the Tyrolean Grey group, according to the expectations. The differences in terms of FA content in the milk fat, instead, were mainly driven by the occurrence of grazing for the low-input group, which led to lower total SFA and higher stearic and vaccenic acid contents (Table 4).

Overview of the Factors Affecting the CH_4 Emissions

Concerning the daily CH_4 production per cow, the results of all equations were affected by the Farming system, with the high-input system resulting in higher CH_4 production values (Table 5). The same applied to parity, with multiparous showing higher CH_4 production than that of primiparous. Both the covariates seasonality and lactation stage were found to affect the CH_4 emissions as well, and this effect was mostly best described by a second degree-polynomial. Moreover, an interaction between farming system and lactation stage was detected for all equations, with a further increase for the emissions of the high-input system (Table 5). The CH_4 emissions per liter milk according to all equations were affected by the farming system, often interacting with the lactation stage, in a way that the emissions per cow, in the high-input system resulting in decreased emissions (Table 6). Accounting for parity in the model improved the model fit for 4 of the 7 equations, and in 3 of the 4 cases, multiparous cows were found to produce lower emissions. Seasonality and lactation stage affected the emissions as well, and interactions between farming system and lactation stage were found to improve the model accuracy for all but one equation (Table 6).

High-Input and Low-Input in Comparison

The investigated equations are based on data sets that investigate diverse dairy farming systems (Table 2); although Engelke et al. (2018) as well as Yan et al. (2006) considered data sets with production parameters from Holstein Friesian dairy cows only kept in investi-

Table 5. Overview of the results of the factors found to affect the CH_4 emissions per cow and day (stepwise forward developed statistical models by means of linear mixed models)[1]

		Source[4]								
Formula[2]	Dependent variable[3]	Intercept	SY	PA	SEA	SEA × SEA	LS	LS × LS	SY × LS	SY × LS × LS
ENG2[5]	*F*	1,247.5	33.8	27.0	3.5	5.1	226.6		12.6	
	P	<0.001	0.001	<0.001	0.064	0.024	<0.001		<0.001	
	E	18.041	3.388	1.614	−0.046	1.0E−03	−0.017		0.006	
	SE	0.6867	0.5826	0.3104	0.0250	4.4E−04	0.0014		0.0018	
RH	*F*	1,275.6	16.2	50.1	31.8	25.3	13.7	22.0	5.1	
	P	<0.001	0.007	<0.001	<0.001	<0.001	<0.001	<0.001	0.024	
	E	400.781	73.943	59.065	3.924	−0.062	0.190	−0.001	0.053	
	SE	16.9252	18.3834	8.3455	0.6961	0.0123	0.0668	0.0002	598.8060	
ENG1	*F*	293.3	13.9	22.0	12.8		24.9	45.2	7.1	16.7
	P	<0.001	0.007	<0.001	<0.001		<0.001	<0.001	0.008	<0.001
	E	373.389	111.401	55.147	0.759		0.862	−0.003	−0.596	0.003
	SE	32.1515	29.8879	11.7638	0.2122		0.1877	0.0006	0.2244	0.0006
MILLS	*F*	1,424.0	16.2	52.0	32.6	26.0	14.4	22.8	5.2	
	P	<0.001	0.007	<0.001	<0.001	<0.001	<0.001	<0.001	0.023	
	E	17.677	3.064	2.487	0.165	−0.003	0.008	0.000	0.005	
	SE	0.7006	0.7608	0.3448	0.0289	0.0005	0.0028	0.0000	0.0022	
NIU	*F*	318.1	16.5	47.6	27.1	21.2	14.8	23.1	5.7	
	P	<0.001	0.007	<0.001	<0.001	<0.001	<0.001	<0.001	0.017	
	E	8.546	3.830	2.983	0.186	−0.003	0.010	−4.3E−05	0.006	
	SE	0.8686	0.9431	0.4324	0.0358	0.0006	0.0034	9.0E−06	0.0027	
YAN1	*F*	2,418.6	16.1	56.9	33.8	27.1	16.2	24.7	5.5	
	P	<0.001	0.007	<0.001	<0.001	<0.001	<0.001	<0.001	0.019	
	E	473.042	60.224	51.493	3.391	−0.054	0.175	−0.001	0.103	
	SE	13.8664	15.0286	6.8253	0.5832	0.0103	0.0556	0.0001	0.0439	
YAN2	*F*	2,018.2	49.8	55.2	23.5	18.3	13.0	13.0	9.5	
	P	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.002	
	E	464.662	114.640	68.221	3.110	−0.049	0.157	−6.2E−04	0.151	
	SE	16.1468	16.2518	9.1809	0.6412	0.0114	0.0642	1.7E−04	0.0489	

[1]Estimates for farming system are shown for high-input versus low-input; estimates for parity are shown for multiparous versus primiparous.

[2]ENG1 = first equation by Engelke et al. (2018); RH = equation by Ramin and Huhtanen (2012); ENG2 = second equation by Engelke et al. (2018); MILLS = equation by Mills et al. (2003); NIU = equation by Niu et al. (2021); YAN1 = first equation by Yan et al. (2006); YAN2 = second equation by Yan et al. (2006).

[3]*F* = *F*-ratio, E = estimated parameter values.

[4]SY = farming system; PA = parity; SEA = seasonality; LS = lactation stage.

[5]Analysis with square root-transformed data.

gational sites and fed with a pre-defined ration, Ramin and Huhtanen (2012), Niu et al. (2021), and Mills et al. (2003) analyzed databases combining data of several studies, containing a variety of breeds, housing systems, and feed rations. All in all, the dairy farming systems described in all these studies could rather be generally regarded as intensive. Therefore, the equations might be more suitable for the high-input scenario described in our study because it is characterized by a high yielding dairy cattle breed as well as by year-round housing in a freestall housing system with silage- and concentrate-based feeding (Table 2). The low-input group provides an additional range of information on how extensive systems behave in comparison to intensive ones.

Generally, higher CH_4 values for the high-input system compared with the low-input system could be observed when focusing on the CH_4 emission produced per day.

The equation ENG2 (Engelke et al., 2018), taking also the FA content into account, showed large differences between low- and high-input system, especially during the first lactation period. Especially toward the end of the lactation period, the low-input group showed remarkably lower CH_4 production compared with the high-input group (Figure 1). For instance, CH_4 emissions were found to be on average 262.1 L/d for the low-input farming system, whereas for the high-input system, values were in the range between 323 and 531 L/d, with a mean value of 422.1 L of CH_4, which gives an average difference between the 2 systems of 160 L of CH_4 per day. In fact, the highest difference between the 2 systems could be observed when applying this formula. This can be explained by the fact that, in addition to DMI, this equation considers milk FA, which, according to Moss et al. (2000), might be a precise CH_4 proxy because microbial activity is directly linked to FA content.

Table 6. Overview of the results of the factors found to affect the CH_4 emissions per kilogram of milk (stepwise forward developed statistical models by means of linear mixed models)[1]

Formula[2]	Dependent variable[3]	Intercept	Source[4]								
			SY	PA	SY × PA	SEA	SEA × SEA	LS	LS × LS	SY × LS	SY × LS × LS
ENG2	*F*	1,405.3	5.0	7.3	10.7	47.3	50.9	26.7			
	P	<0.001	0.034	0.007	0.001	<0.001	<0.001	<0.001			
	E	12.752	2.035	1.480	−1.586	−0.156	0.003	0.004			
	SE	0.5268	0.6433	0.3228	0.4844	0.0227	0.0004	0.0008			
RH[5]	*F*	4,903.2	5.6	4.5		11.6		1,151.6		73.7	
	P	<0.001	0.045	0.034		<0.001		<0.001		<0.001	
	E	1.262	−0.052	−0.015		5.0E−04		9.2E−04		−3.6E−04	
	SE	0.0216	0.0218	0.0073		1.5E−04		3.4E−05		4.2E−05	
ENG1[6]	*F*	3,184.7	1.115			4.147		119.483	34.0	13.5	10.5
	P	<0.001	0.299			0.043		<0.001	<0.001	<0.001	0.001
	E	3.7997	−0.1298			0.0021		0.0088	−1.6E−05	−0.004	1.1E−05
	SE	0.0925	0.1230			0.0010		0.0010	0.0000	0.0012	0.0000
MILLS[4]	*F*	538.9	2.7			9.4		1,338.7		122.5	
	P	<0.001	0.142			0.002		<0.001		<0.001	
	E	0.234	−0.020			2.2E−04		5.1E−04		−2.4E−04	
	SE	0.0115	0.0119			7.2E−05		1.7E−05		2.2E−05	
NIU[4]	*F*	1,224.2	0.2			14.6		358.0		27.1	
	P	<0.001	0.672			<0.001		<0.001		<0.001	
	E	0.155	0.004			3.0E−04		3.6E−04		−1.6E−04	
	SE	0.0061	0.0083			8.0E−05		2.3E−05		3.0E−05	
YAN1[4]	*F*	4,510.5	15.4	10.6		6.3		1,209.2		71.4	
	P	<0.001	0.002	0.001		0.012		<0.001		<0.001	
	E	1.320	−0.072	−0.024		3.7E−04		9.3E−04		−3.6E−04	
	SE	0.0220	0.0183	0.0073		1.5E−04		3.4E−05		4.2E−05	
YAN2[4]	*F*	4,850.2	3.1	5.6		4.5	7.9	1,252.1		65.0	
	P	<0.001	0.095	0.018		0.034	0.005	<0.001		<0.001	
	E	1.327	−0.036	−0.018		−1.4E−03	3.2E−05	9.7E−04		−3.5E−04	
	SE	0.0221	0.0201	0.0075		6.5E−04	1.1E−05	3.5E−05		4.3E−05	

[1]Estimates for farming system are shown for high-input versus low-input; estimates for parity are shown for multiparous versus primiparous.

[2]ENG1 = first equation by Engelke et al. (2018); RH = equation by Ramin and Huhtanen (2012); ENG2 = second equation by Engelke et al. (2018); MILLS = equation by Mills et al. (2003); NIU = equation by Niu et al. (2021); YAN1 = first equation by Yan et al. (2006); YAN2 = second equation by Yan et al. (2006).

[3]*F* = *F*-ratio, E = estimated parameter values.

[4]SY = farming system; PA = parity; SEA = seasonality; LS = lactation stage.

[5]Analysis with logarithm (base 10)-transformed data.

[6]Analysis with square root-transformed data.

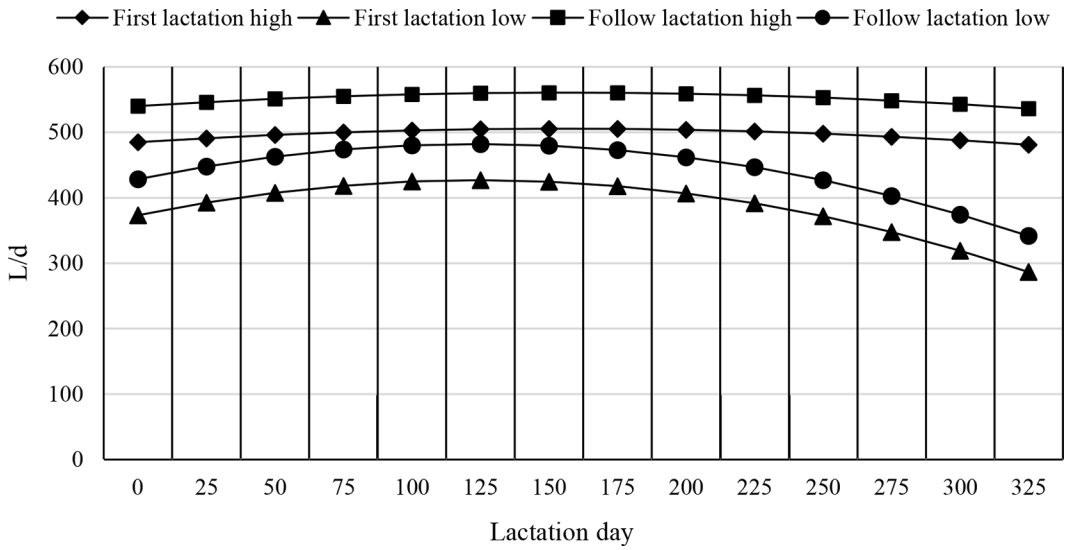


Figure 1. Predicted values of CH_4 production in liters per day according to the second equation by Engelke et al. (2018), depending on the farming system, parity, and lactation stage. Concerning seasonality, the results are referred to the first week of the year.

For ENG2, which considers ECM as a calculation factor, results were slightly different, indicating values of approximately 524.4 L for the concentrate-based system and 414.0 L of daily CH_4 production for the pasture-based system. As a result, variation between the 2 systems was slightly lower when applying this second equation, which might be explained using different variables within the formulas (DMI for the first and ECM for the second one). However, as Hristov et al. (2013) have reported, ECM might be well reflecting the DMI of cows, and at the same time, be a more easily available parameter in practice (e.g., via official milk recording scheme). Thus, calculations based on ECM should give similar values as those based on DMI with slight differences between the equations, as highlighted by our results. Similarly, Ramin and Huhtanen (2012) yielded an estimated CH_4 value for intensive system of 524.6 L and between 377 and 471 L for the extensive system, which results in an average difference of 93 L of CH_4 per day. Again, this equation is based on DMI, which accounts for 52% to 64% of daily CH_4 production when cattle is fed ad libitum (Knapp et al., 2014).

The 2 equations of Yan et al. (2006) showed high values for both, low- and high-input structures. In fact, results for the first equation were on average 501.03 L for the extensive system and 577.99 L of CH_4 for the intensive system, whereas calculations for the second formula resulted in average emissions of 501.73 and 640.86 L for the low and high-input system, respectively. Furthermore, high differences between the 2 systems could be observed when applying the YAN2, whereas the lowest discrepancy between the 2 systems resulted when applying YAN1.

In contrast to these findings, calculation showed significantly different results when considering the amount of CH_4 emitted per kilogram of milk produced as a calculation factor instead of daily CH_4 production. Average values were higher for the low-input group (21.86 L of CH_4 per L of milk), whereas high-input values were slightly lower (17.88 L of CH_4 per L of milk) when applying the ENG2. The difference between the high and low group was quantified at −3.98 L as an average value, with higher CH_4 production for the extensive system (Figure 2). Similar results were observed for all other equations examined within our study. The highest discrepancy between the 2 systems was found when applying the equation of YAN2 with the low-input group producing 7.94 L of CH_4 more than the high-input group. As for the previous calculations with liters of CH_4 per day, emission values, also in this case, were generally high for the equations of Yan et al. (2006), with a CH_4 production of 28.55 and 29.61 L of CH_4 for the extensive farming group and 20.60 and 23.40 L for the intensive farming group, for YAN1 and YAN2, respectively.

The same difference in CH_4 production (−6.12 L of CH_4 per L of milk) could be found when adopting the formulas RH (Ramin and Huhtanen, 2012) and YAN2 (Yan et al., 2006). Nevertheless, RH showed significantly lower amounts of emitted CH_4 for both low and high-input farming with values of 24.94 and 18.82 L of CH_4, respectively, which are similar results to those obtained by using ENG2 (Engelke et al.,

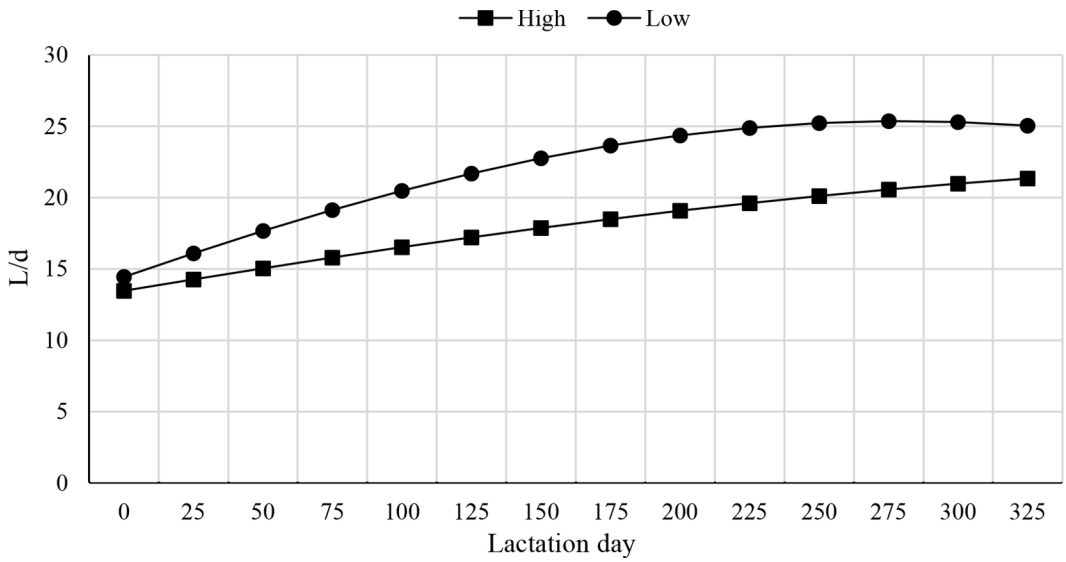


Figure 2. Predicted values of CH_4 production in liters per kilogram of milk according to the second equation by Engelke et al. (2018), depending on the farming system and lactation stage. The analysis was performed with square root-transformed data; back-transformed functions are shown. Concerning seasonality, the results are referred to the first week of the year.

2018). This can be explained with the higher milk production level from the Simmental cattle compared with the Tyrolean Grey cattle. Indeed, as already reported by several studies, when increasing milk yield, CH_4 production per liter of milk decreases as a logical consequence, due to a dilutive effect (Jiao et al., 2014; O'Brien et al., 2014; Lorenz et al., 2019). In fact, Jiao et al. (2014) has shown that increasing the amount of concentrate in the ration (which leads to higher milk productivity) might lead to reduced CH_4 production per unit of milk, but daily CH_4 amounts would remain unaffected.

Only when using the ENG2 equation by Engelke et al. (2018), CH_4 production was 14.15 L per kilogram of milk for the low-input system and 15.39 L for the high-input system on average. The difference (Δ 1.24 L of CH_4 per kilogram of milk) in favor of the low-input system is explainable by the higher content of milk fat (4.7% vs. 4.2%) as well as the higher content of SFA (65.1 g/100 g of total FA vs. 61.5 g/100 g of total FA) in Simmental milk.

The equations MILLS (Mills et al., 2003) as well as NIU (Niu et al. (2021) consider the production of CH_4 in megajoules per day (MJ/d) and megajoules per kilogram of milk produced (MJ/kg of milk), making use of the DMI as variable for the equations.

Generally, much lower values were obtained with the formula NIU (Niu et al. (2021), with an average 10.11 MJ of CH_4 produced by the low-input group and 14.99 MJ produced by the high-input group, resulting in a difference of 4.88 MJ/d (Figure 3). In contrast to that, the difference between the 2 systems was on average 3.87 for the equation of MILLS (Mills et al., 2003). Moreover, generally higher CH_4 amounts could be detected with the use of the formula reported in Mills et al. (2003), ranging from 18.98 to 22.86 MJ/d for the low- and the high-input system, respectively. When calculating the produced MJ of CH_4 per kilogram of milk, the low-input system produced more CH_4 (0.32 and 0.21 MJ of CH_4 per kilogram of milk on average), than the high-input system (0.26 and 0.19 MJ of CH_4 per kilogram of milk on average; Figure 4). As described previously, due to dilution effect, CH_4 production is proportionally smaller for the high-input group when counting the emissions per kilogram of milk instead of counting absolute emissions.

Effect of Parity and Lactation Stage on CH_4 Emissions

When comparing primiparous with multiparous cows, as well as the lactation stage, lowest emissions were generally observed during the first lactation when compared to the following lactations. This can be explained by the fact that in primiparous cows, milk yield (Shanks et al., 1981) along with DMI, is lower than in multiparous cows. The smallest difference between first and following lactations was shown when applying the first equation (YAN1) published by Yan et al. (2006), with a difference in overall CH_4 emission of 51.5 L of CH_4 per day. Again, highest difference (68.2 L of CH_4 per d) could be observed for the second formula (YAN2) of Yan et al. (2006), which, however, generally showed highest CH_4 emission values for the analyzed

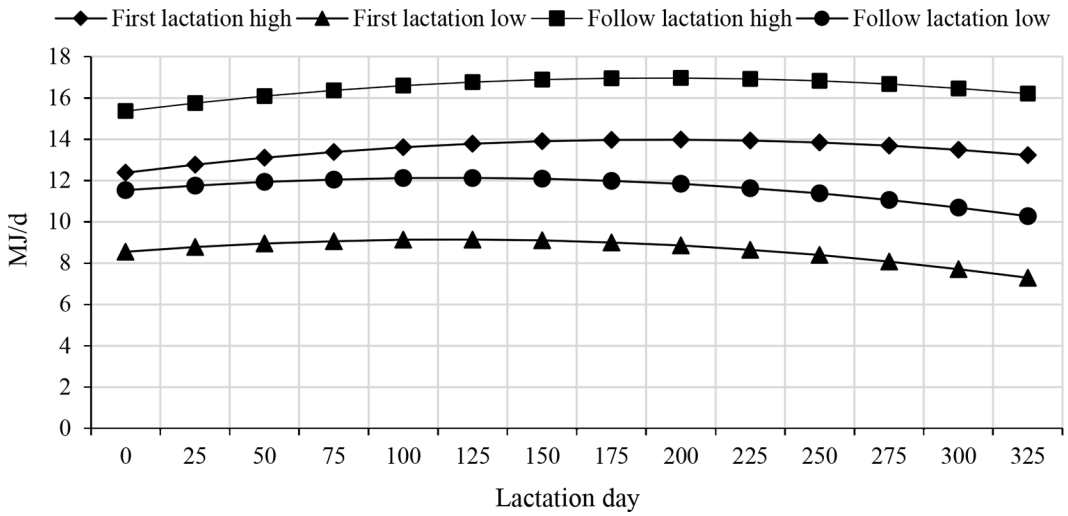


Figure 3. Predicted values of CH_4 production in megajoules per day according to the equation by Niu et al. (2021), depending on the farming system, parity, and lactation stage. Concerning seasonality, the results are referred to the first week of the year.

system. This might be explainable by the fact that this equation, in addition to DMI, includes BW as variable. However, it has been shown elsewhere that BW and CH_4 production do not have any significant correlation and, hence, CH_4 emission should not be influenced by BW (Breider et al., 2019).

In addition to that, when assessing CH_4 values obtained per day, a clear trend of lowest emissions during the beginning and the end of the lactation period and highest emissions toward the lactation peak could be observed, which is in line with the findings of Veerkamp and Thompson (1999). Solely, the results based on the equation (ENG2) of Engelke et al. (2018) showed a different pattern with highest CH_4 values at the beginning of the lactation period and decreasing CH_4 production with continuous lactation, which might be explained by the parameter of milk FA. In fact, Vanrobays et al. (2016) has shown that correlations between milk FA content and CH_4 production vary significantly during the lactation period.

When calculating CH_4 emissions per kilogram of milk, CH_4 values are generally highest toward the end of the lactation, whereas lowest values were observed at the beginning of the lactation period. This finding is

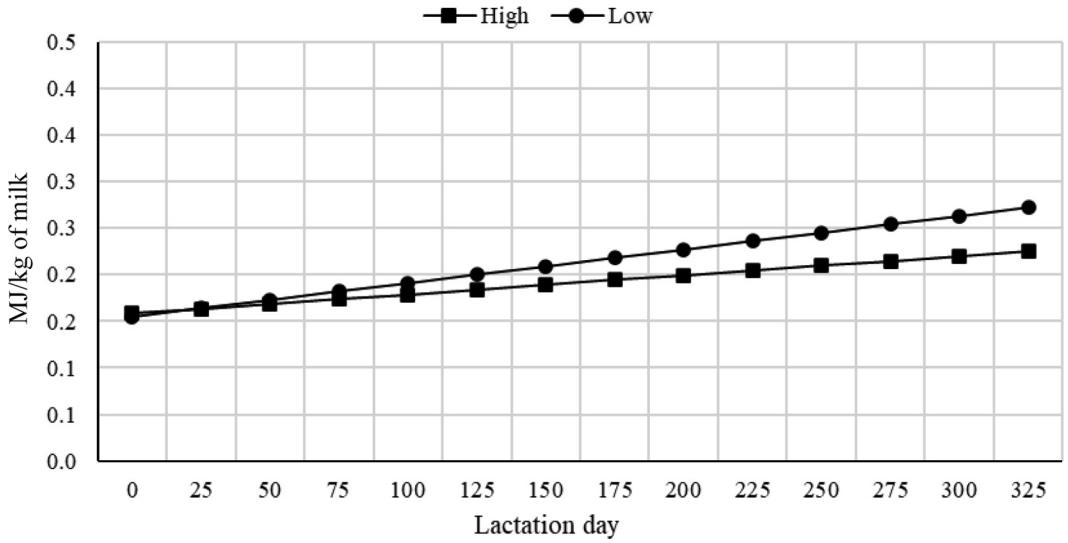


Figure 4. Predicted values of CH_4 production in megajoules per kilogram of milk according to the equation by Niu et al. (2021), depending on the farming system, parity, and lactation stage. The analysis was performed with logarithm (base 10)-transformed data; back-transformed functions are shown. Concerning seasonality, the results are referred to the first week of the year.

Table 7. Pearson correlation between the results of the equations used to estimate the CH_4 production per cow (CH_4 amount/animal per day)

	Formula						
Formula[1]	ENG1	RH	ENG2	MILLS	NIU	YAN1	YAN2
ENG1	1.000	0.635**	0.695**	0.635**	0.640**	0.630**	0.652**
RH		1.000	0.801**	0.999**	0.999**	0.987**	0.941**
ENG2			1.000	0.804**	0.805**	0.804**	0.751**
MILLS				1.000	0.998**	0.993**	0.94**
NIU					1.000	0.982**	0.945**
YAN1						1.000	0.927**
YAN2							1.000

[1]ENG1 = first equation by Engelke et al. (2018); RH = equation by Ramin and Huhtanen (2012); ENG2 = second equation by Engelke et al. (2018); MILLS = equation by Mills et al. (2003); NIU = equation by Niu et al. (2021); YAN1 = first equation by Yan et al. (2006); YAN2 = second equation by Yan et al. (2006).
**$P < 0.01$.

perfectly in line with the data provided by Veerkamp and Thompson (1999), who revealed that DMI does not rise as fast as milk yield at the beginning of the lactation period, whereas it continues to increase together with milk yield in the following stages of the lactation period. At the end of the lactation, milk productivity decreases faster than feed intake, which explains the highest CH_4 values per kilogram of milk during this period.

No significant difference between first and following lactations could be observed for the equations ENG2, MILLS, and NIU. This could denote a low ability of these equations in illustrating differences between lactations.

Correlations Between Equations

Strong correlations could be observed for all equations when examining CH_4 emissions as liters per day. In fact, correlation coefficients (r) varied between 0.630 and 0.999 (Table 7). Lower r values were achieved by the results of ENG1 with those of all other equations, which points out the differentiating role played by the inclusion of the FA percentage in the equation. In contrast, for those calculations based on CH_4 production per kilogram of milk, results were more diverse, indicating stronger differences between the results of different equations (Table 8). Although positive correlations were detected, r values range from 0.021 to 0.999 with largely varying *P*-values. Nonsignificant correlations were observed between equations MILLS and ENG1 (0.041), RH and ENG1 (0.046), as well as between YAN1 and ENG1 (0.021; Table 8). In contrast, the formula ENG2 shows strong correlations to other equations based on DMI only and could, therefore, be used as an alternative formula for evaluating CH_4 emissions (Table 8).

CONCLUSIONS

In the present study, CH_4 emissions of 2 different dairy production systems typical for small-scale mountain dairy farms were estimated by previously developed equations, which consider routinely collected

Table 8. Pearson correlation between the results of the equations used to estimate the CH_4 production per kilogram of milk (CH_4 amount/kg of milk)

	Formula						
Formula[1]	ENG1	RH	ENG2	MILLS	NIU	YAN1	YAN2
ENG1	1.000	0.046^{NS}	0.321**	0.041^{NS}	0.150**	0.021^{NS}	0.089*
RH		1.000	0.809**	1,000**	0.883**	0.993**	0.976**
ENG2			1.000	0.808**	0.787**	0.794**	0.783**
MILLS				1.000	0.875**	0.995**	0.976**
NIU					1.000	0.827**	0.860**
YAN1						1.000	0.969**
YAN2							1.000

[1]ENG1 = first equation by Engelke et al. (2018); RH = equation by Ramin and Huhtanen (2012); ENG2 = second equation by Engelke et al. (2018); MILLS = equation by Mills et al. (2003); NIU = equation by Niu et al. (2021); YAN1 = first equation by Yan et al. (2006); YAN2 = second equation by Yan et al. (2006).
*$P < 0.05$; **$P < 0.01$; NS ($P \geq 0.05$).

production parameters, such as milk FA and milk solids. No direct measurements (e.g., respiration chamber) for CH_4 emissions were performed to calculate estimation accuracy of selected equations. Nevertheless, the results clearly demonstrated the potential of using such equations under field conditions on a large scale using routinely collected parameters for quantifying CH_4 emissions to compare different dairy production systems, in terms of their relevance for greenhouse gas emissions in a cost-effective way. The latter could open new perspectives for breeding purpose and management decisions. Consequently, results of this study should help in providing useful information in the debate on sustainable milk production and the development of future climate friendlier production systems in regions where the on-farm production of concentrates is not possible due to climatic and topographic constraints (e.g., mountain area) and, thus, are mainly imported from other regions.

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